Uber / Taxi Trip Data Analysis Using Apache Spark

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Academic Year: 2026-2027

PROBLEM STATEMENT

Taxi trip datasets are large and complex with trip times, distances, fares, and locations

Manual analysis is difficult and time-consuming

Need for a system to process data, identify patterns, calculate revenue stats, and visualize results

Goal: Create an easy-to-use dashboard for insights

PROJECT OBJECTIVES

Process taxi trip data using Apache Spark

Clean invalid and duplicate records

Calculate total trips, revenue,
average fare, and distance

Analyze trips by hour, date,
and pickup location

Display insights with interactive charts

Enable filtering and exporting
of trip records

TECHNOLOGIES USED

React

Frontend dashboard interface

TypeScript

Frontend programming language

Python Flask

Backend REST API server

Apache Spark / PySpark

Data cleaning and analysis engine

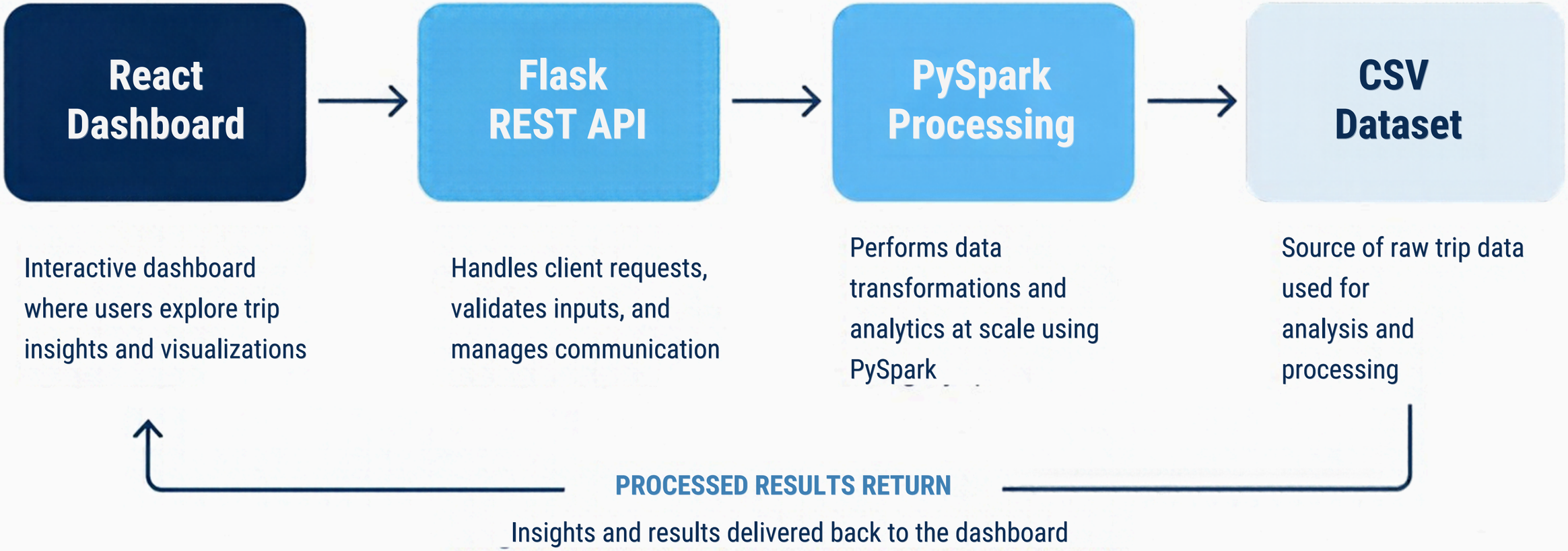
Chart.js

Visualization of data with charts

CSV

Dataset input and storage format

SYSTEM ARCHITECTURE



DATA ANALYSIS MODULE

From raw trip data to meaningful insights

01 **Data Loading**
Load trip data from CSV files

02 **Data Cleaning**
Handle missing, duplicate, and invalid records

03 **Trip Summary**
Compute total trips, revenue, average fare, and distance

04 **Hourly Analysis**
Analyze trips by hour of day

05 **Pickup Location Analysis**
Compare trip counts by pickup location

06 **Daily Revenue Analysis**
Analyze revenue trends by date

07 **Filtering and Export**
Apply user filters and export results to CSV

RESULTS AND OUTCOME

50,000

SOURCE ROWS

42,538

VALID COMPLETED TRIPS AFTER CLEANING

$679,678.85

TOTAL REVENUE

$15.98

AVERAGE FARE

7.01 km

AVERAGE TRIP DISTANCE

These values reflect this dataset only, not universal Uber stats

CONCLUSION AND FUTURE SCOPE

CONCLUSION

Developed a web-based taxi trip analytics dashboard

Used PySpark for data cleaning and aggregation

Connected frontend React with Flask backend

Presented trip and revenue data via interactive charts

FUTURE SCOPE

Integrate Cassandra for persistent storage

Deploy Spark on distributed cluster

Add map-based visualizations

Implement demand prediction with machine learning

Support automated dataset updates

THANK YOU

Explore the live dashboard

https://uber-taxi-trip-data-analysis--24modanip.replit.app/